

Mutual Fund Analytics & Investor Intelligence Dashboard

Bluestock Fintech Data Analyst Capstone Project

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Tools Used:

- Python
- Pandas
- NumPy
- Matplotlib
- SQL
- Power BI

Executive Summary

The Mutual Fund Analytics & Investor Intelligence Project was developed to analyze mutual fund performance, investor behavior, portfolio risk, and SIP trends using a combination of Python, SQL, and Power BI.

The project involved building a complete data analytics pipeline beginning with data ingestion and cleaning, followed by exploratory data analysis, performance scoring, advanced risk analytics, and dashboard development.

The final solution provides insights into:

- Mutual fund performance
- Risk-adjusted returns
- Investor demographics
- SIP investment behavior
- Portfolio concentration
- Fund recommendations

The dashboard enables stakeholders to identify high-performing funds, understand investor trends, and support data-driven investment decisions.

Project Objectives

Business Objectives:

The key objectives of the project were:

1. Analyze mutual fund industry trends.
2. Measure fund performance using multiple financial metrics.
3. Create a composite fund scoring model.
4. Study investor demographics and behavior.
5. Analyze SIP continuity and investor retention.
6. Perform advanced risk analysis using VaR and CVaR.
7. Build interactive Power BI dashboards.
8. Generate actionable business insights.

Data Sources

Data Sources

The project utilized multiple datasets covering mutual fund operations and investor activity.

Datasets Used:

Fund Master

Contains scheme-level information.

NAV History

Historical Net Asset Value records.

Investor Transactions

Investor purchases and SIP transactions.

Portfolio Holdings

Fund portfolio composition and sector allocations.

Industry AUM Data

Assets Under Management across the industry.

Fund Performance Metrics

Returns, Sharpe Ratio, Alpha, Beta and Drawdown metrics.

Total Records Processed

More than 150,000 records were processed during the project lifecycle.

ETL Architecture

ETL Design

The project followed a structured ETL pipeline.

Extract

- CSV datasets imported using Python.
- Data loaded into Pandas DataFrames.

Transform

- Missing value treatment.
- Data type corrections.
- Duplicate removal.
- Outlier validation.
- Metric calculations.

Load

- Clean datasets stored in Processed folder.
- Final outputs exported for dashboard consumption.

Tools Used

- Python
- Pandas
- NumPy
- SQL

Data Cleaning & Quality

Data Cleaning Activities

The following cleaning operations were performed:

Missing Value Handling

- Null values identified and treated.
- Missing numerical values imputed where appropriate.
-

Duplicate Removal

Duplicate records removed from transaction and NAV datasets.

Data Type Corrections

- Dates converted to datetime format.
- Numeric columns converted appropriately.
-

Validation Checks

- Invalid transaction amounts identified.
- Incorrect category values corrected.
-

Outcome

The final datasets achieved high consistency and reliability for analysis.

Exploratory Data Analysis

Key EDA Findings

Investor Demographics

- Majority of investors belong to middle-age categories.
- Male investors represented a larger share of the dataset.

Geographic Analysis

- Tier 1 cities contributed the highest investor participation.
- Major urban centers dominated investment activity.

Fund Categories

- Equity funds represented the largest category.
- Hybrid and Debt funds showed stable participation.

Performance Analytics

Fund Performance Evaluation

Multiple financial metrics were analyzed:

Metrics Used

- 3-Year Return
- Sharpe Ratio
- Alpha
- Beta
- Maximum Drawdown
- Expense Ratio
-

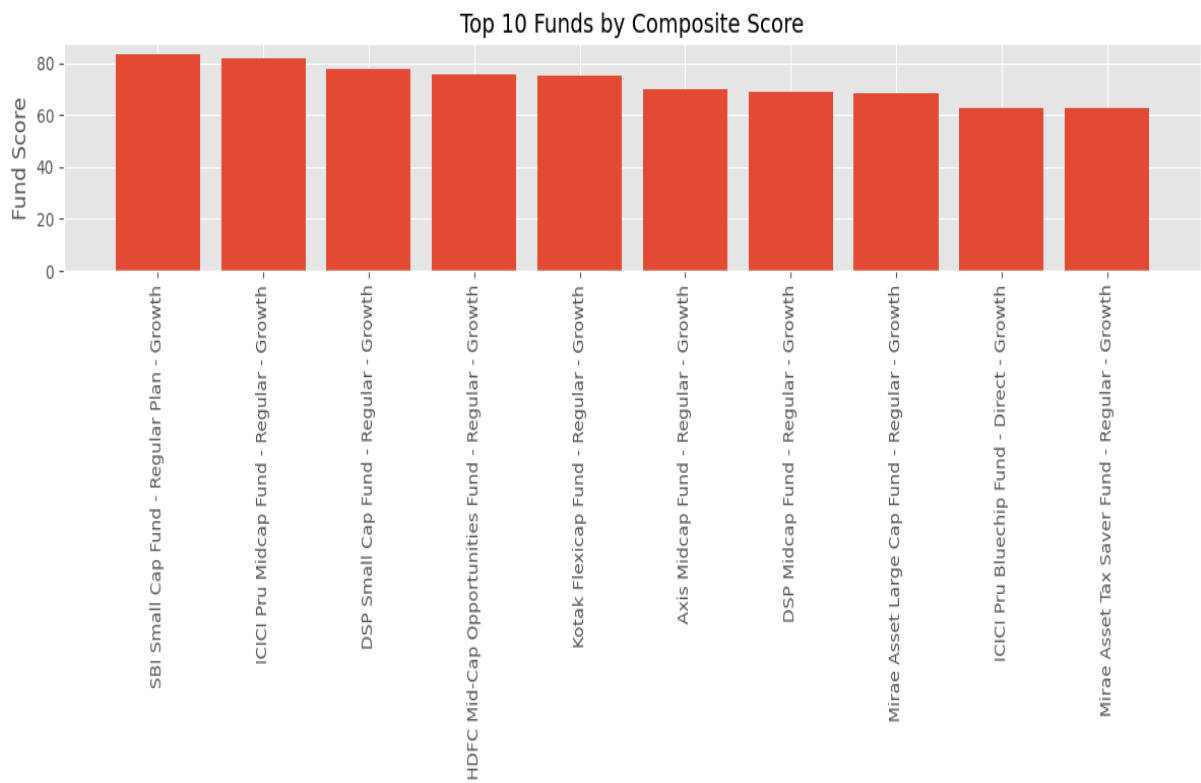
Fund Score Model

A weighted scoring framework was created to rank funds.

The model combines:

- Return Performance
- Risk Adjustment
- Drawdown Protection
- Cost Efficiency

Top 10 fund Scorecard



Advanced Analytics

Risk Analysis

Value at Risk (VaR)

Historical VaR was calculated using the 95% confidence level.

Purpose:

Estimate potential downside risk under normal market conditions.

Conditional Value at Risk (CVaR)

Measures expected loss beyond the VaR threshold.

Findings

Several schemes exhibited significantly higher downside risk compared to peers.

This information helps investors understand tail-risk exposure.

Rolling Sharpe Analysis

Rolling Sharpe Ratio

A 90-day rolling Sharpe Ratio was calculated.

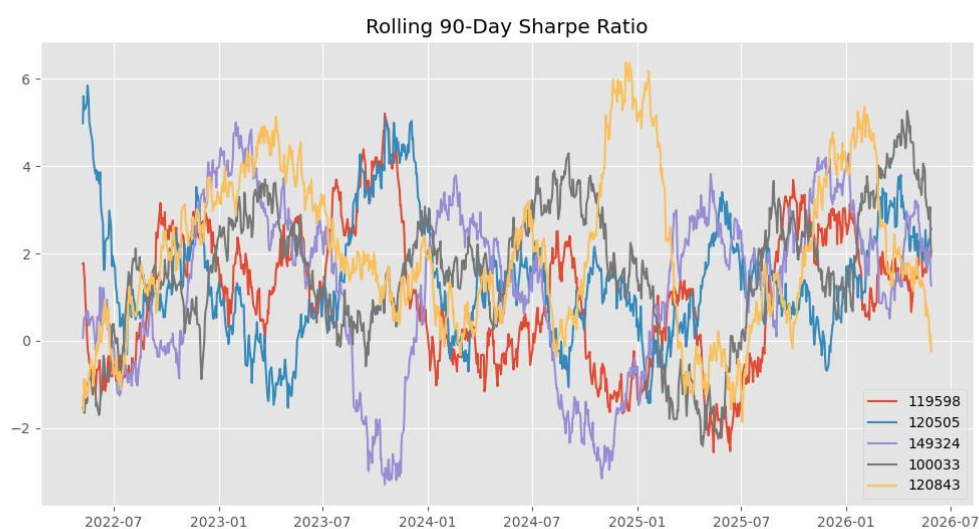
Formula:

$$\text{Sharpe Ratio} = \text{Mean Return} / \text{Standard Deviation} \times \sqrt{252}$$

Benefits

- Measures consistency of performance.
- Tracks risk-adjusted returns over time.
- Identifies periods of fund strength and weakness.

Rolling 90- days Sharpe Ratio



Investor Analytics

Investor Behavior Analysis

Investor transactions were analyzed to identify behavior patterns.

Key Findings

- Tier 1 cities generated highest participation.
- Digital payment modes dominated transactions.
- SIP investors showed stronger long-term engagement.
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Cohort Analysis

Investors were grouped based on first investment year.

Metrics analyzed:

- Average investment amount
- Total invested capital
- Preferred fund categories

SIP Continuity Analysis

SIP Continuity Study

The continuity of SIP investments was analyzed.

Methodology

Investors with at least six SIP transactions were evaluated.

Average gap between transactions was calculated.

Risk Classification

Healthy:

- Gap $\leq$ 35 Days

At Risk:

- Gap $>$ 35 Days

Findings

A large proportion of investors showed irregular SIP patterns indicating possible churn risk.

Dashboard Overview

Power BI Dashboard

The final dashboard consists of four interactive pages:

Page 1

Industry Overview

Page 2

Fund Performance

Page 3

Investor Analytics

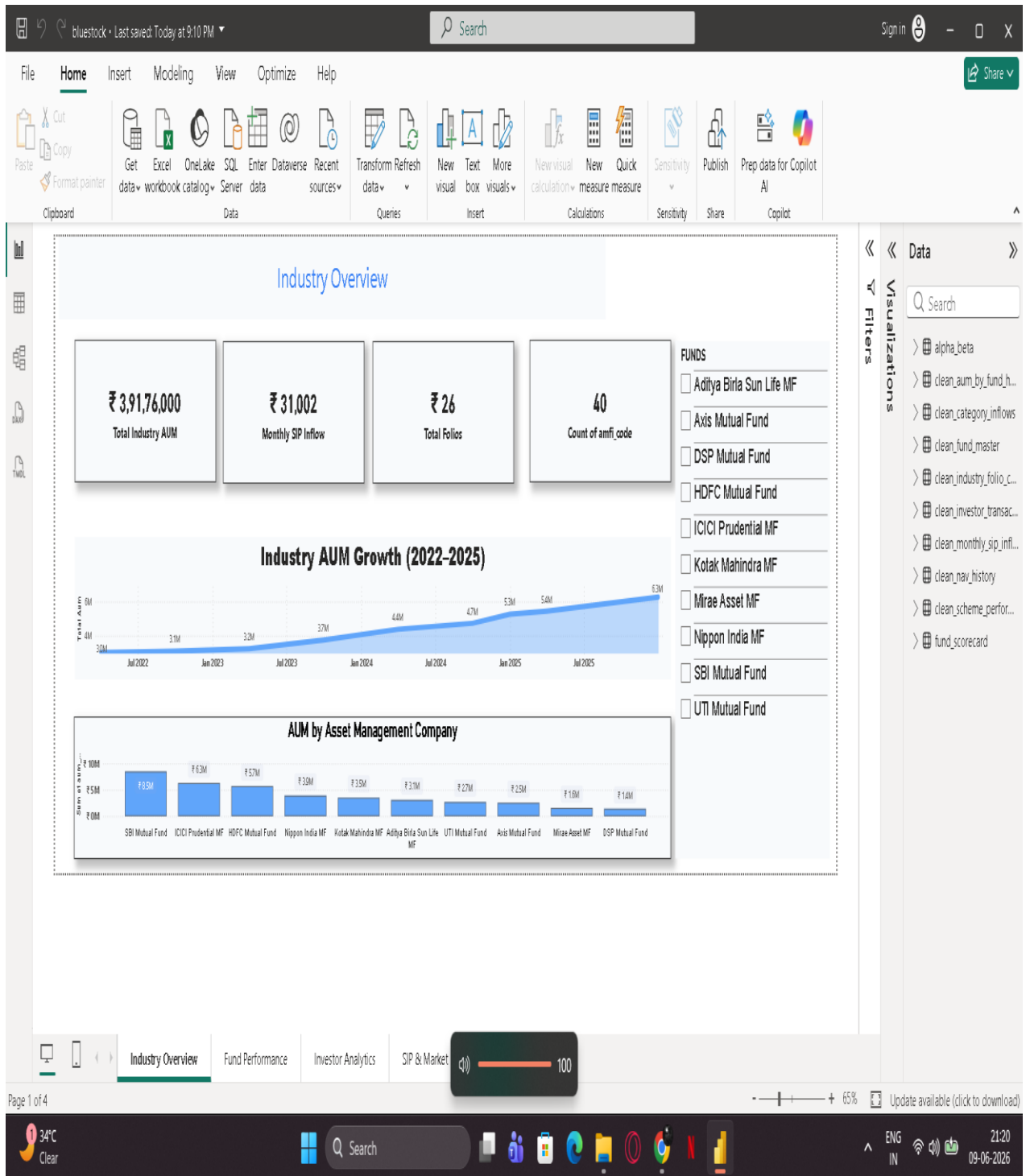
Page 4

SIP Market Trends

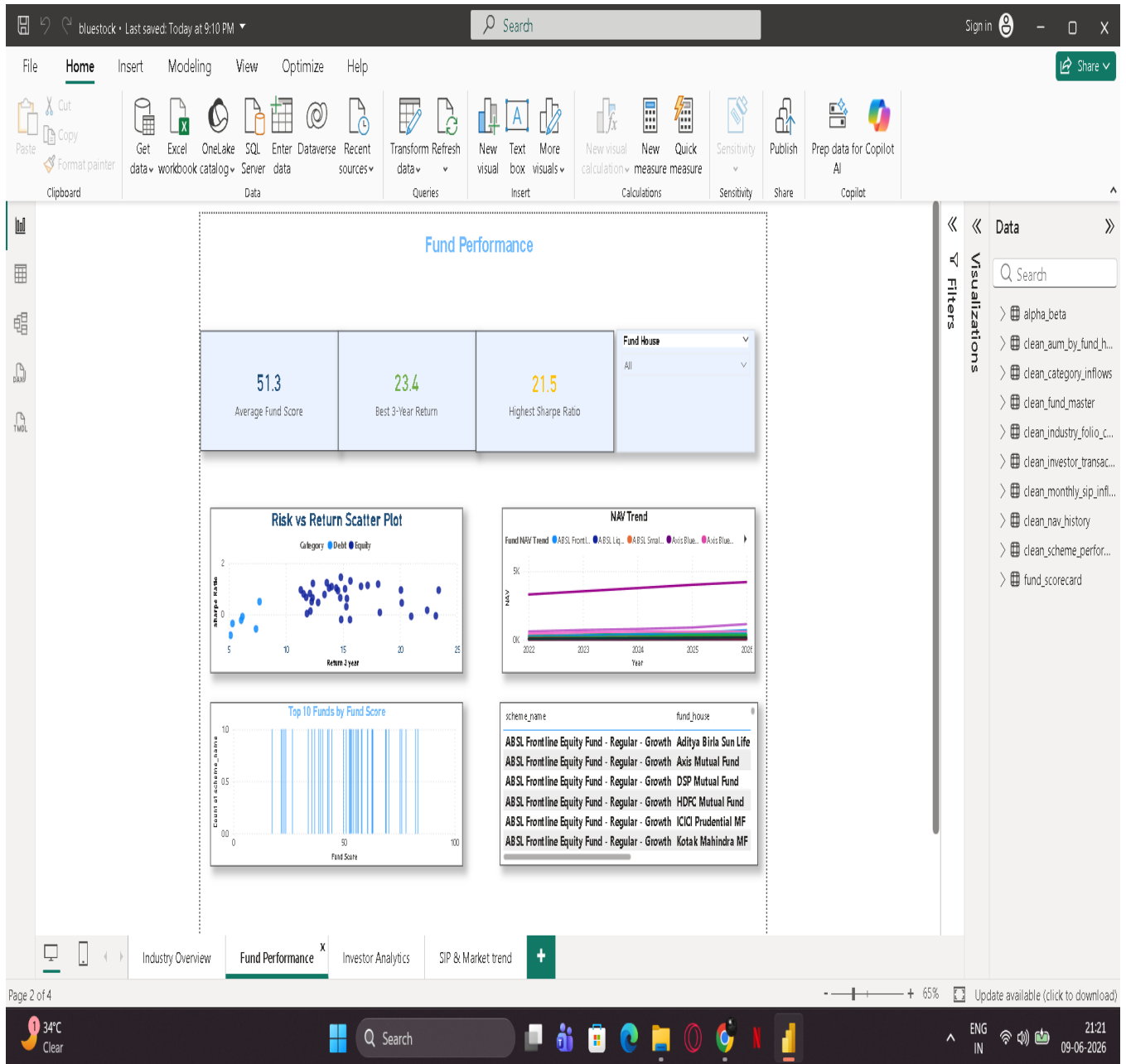
Features:

- KPI Cards
- Interactive Filters
- Dynamic Visualizations
- Fund Ranking System

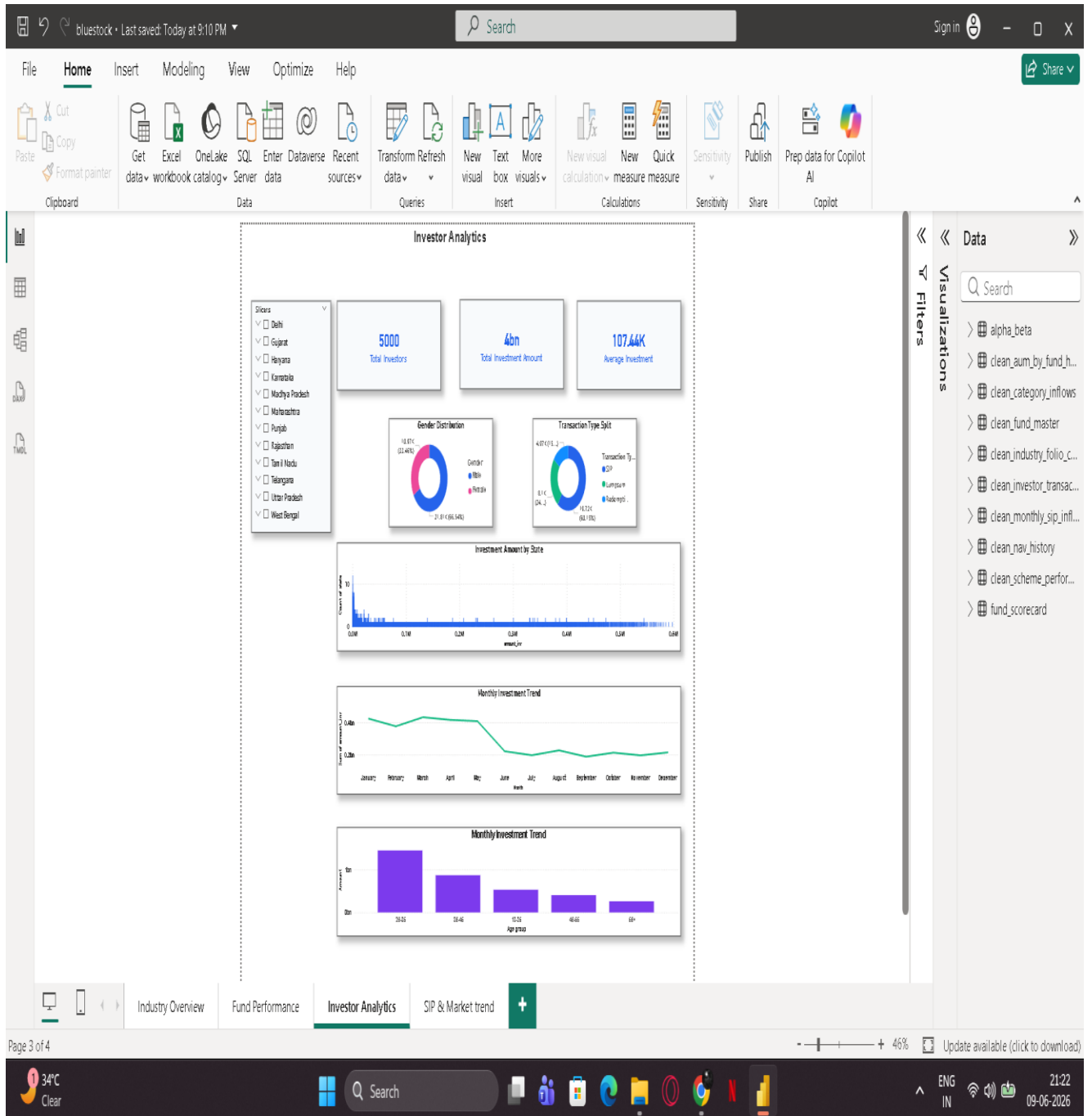
Industry Overview Dashboard



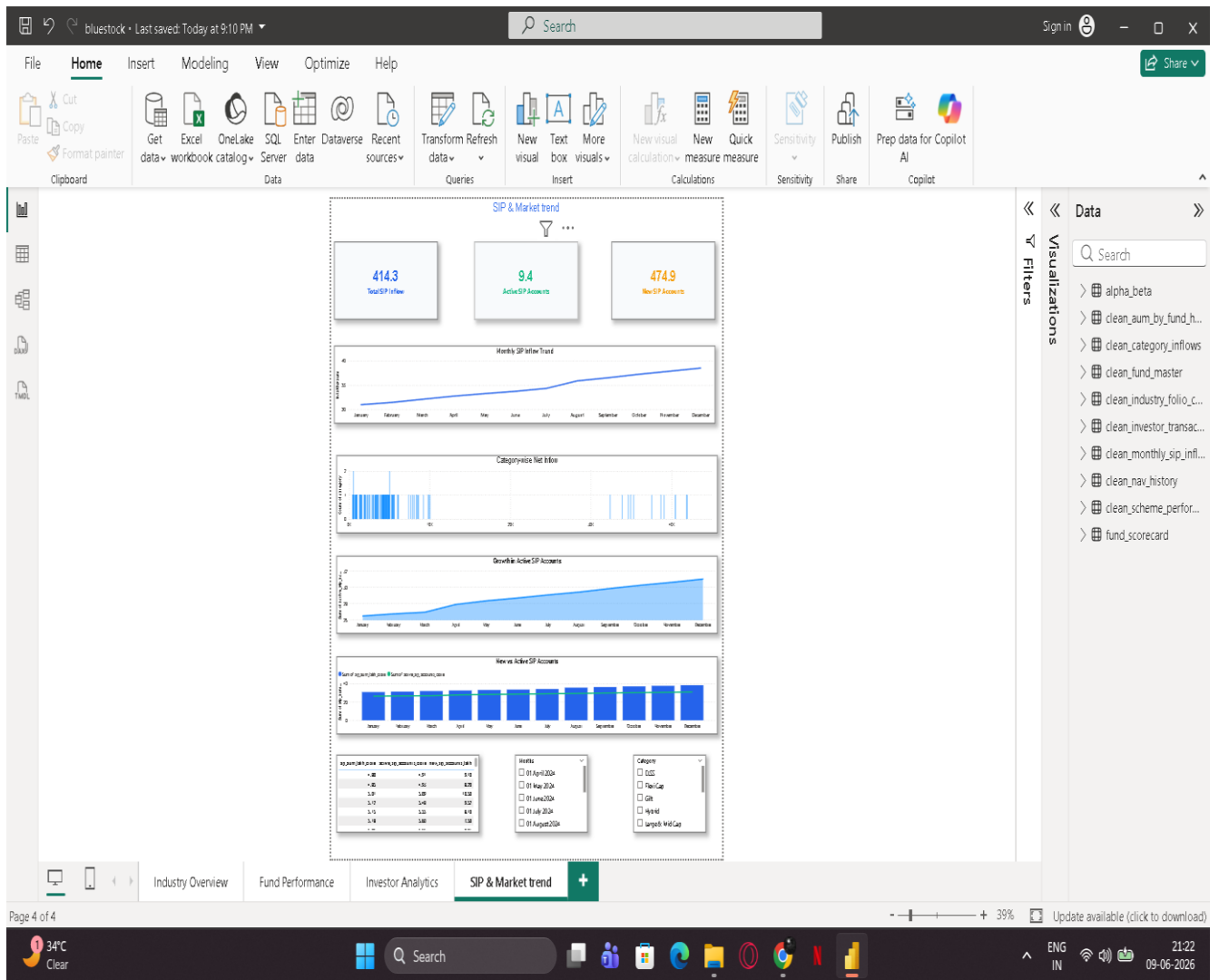
Fund Performance Dashboard



Investor Analytics Dashboard



SIP Market Trends Dashboard



Key Business Insights

Key Business Insights

1. Funds with higher Sharpe Ratios consistently achieved stronger risk-adjusted performance.
2. Equity funds generated the highest long-term returns.
3. Tier 1 cities accounted for the majority of investment activity.
4. Several investors displayed SIP gaps exceeding 35 days, indicating retention risk.
5. Portfolio concentration varied significantly across funds, highlighting diversification differences.

Limitations

Project Limitations

- Dataset represents a simulated environment.
- Limited historical coverage for some schemes.
- Market events outside the dataset period were not considered.
- Fund recommendation model is rule-based and not machine learning driven.

Recommendations

Recommendations

For Asset Management Companies

- Improve SIP retention campaigns.
- Promote diversified portfolios.
- Enhance investor education programs.

For Investors

- Focus on risk-adjusted returns rather than absolute returns.
- Monitor portfolio concentration.
- Maintain SIP discipline for long-term wealth creation.

Conclusion

Conclusion

The Mutual Fund Analytics & Investor Intelligence Project successfully transformed raw financial and investor data into actionable insights.

The project demonstrates practical application of:

- Data Engineering
- Data Analysis
- Financial Analytics
- Dashboard Development
- Business Intelligence

The solution provides a strong foundation for informed investment decision-making and portfolio monitoring.

Thank You

Mutual Fund Analytics & Investor Intelligence Dashboard

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